DefMamba: Deformable Visual State Space Model

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Abstract

Visual state space models are efficient, but they still rely heavily on fixed scan patterns that flatten a 2D feature map into a rigid 1D sequence. This weakens spatial adjacency, breaks object structure, and limits content awareness. We present DefMamba, a hierarchical visual backbone built on a Deformable State Space Model (DSSM) that learns both where to sample and how to order tokens before selective scanning. The proposed deformable branch predicts bounded 2D point offsets for bilinear feature sampling together with token-index offsets that induce a content-adaptive sequence through differentiable sorting. To keep optimization stable, DefMamba retains standard forward and backward scan branches and averages them with the deformable branch. Across three vision benchmarks, DefMamba is consistently competitive. On ImageNet-1K, DefMamba-T/S/B achieve 78.6%, 83.5%, and 84.2% top-1 accuracy at 1.2G, 4.8G, and 8.5G FLOPs. With Mask R-CNN on COCO, DefMamba-S reaches 47.5 box AP and 42.8 mask AP. With UPerNet on ADE20K, DefMamba-S obtains 48.8 mIoU under single-scale testing and 49.6 under multi-scale testing. Ablations show that deforming either points or token order helps, while combining both delivers the largest gain.

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1. Introduction

State space models (SSMs) have re-emerged as an efficient alternative to attention for long-sequence modeling, progressing from structured SSMs such as S4 [7] to selective SSMs such as Mamba [6]. Their linear-time complexity makes them attractive for visual backbones, and recent models including ViM [33], VMamba [15], LocalMamba [13], MSVMamba [21], GrootVL [27], and QuadMamba [28] confirm that selective scan can be competitive on classification and dense prediction. However, a central mismatch remains: images are two-dimensional, while selective scan consumes an ordered one-dimensional sequence.

Most visual SSMs address this mismatch through handcrafted scanning rules. Rasterized sequences preserve implementation simplicity but break local spatial adjacency.

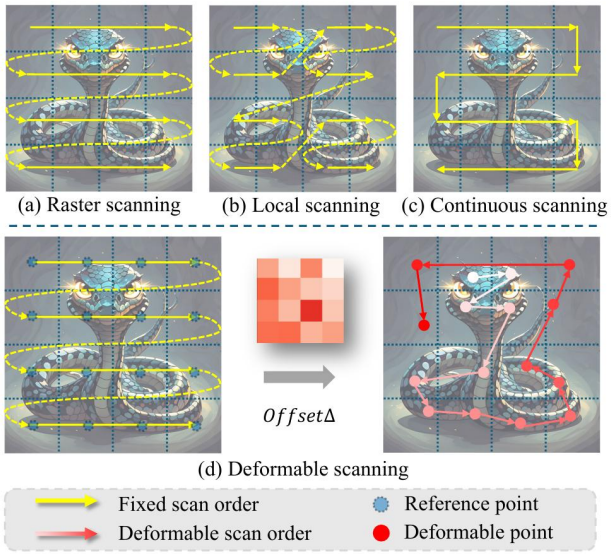

Figure 1. Comparison of different scan strategies. Fixed raster, local, and continuous scans follow hand-crafted paths, while DefMamba predicts content-adaptive point and token-index offsets to better align the 1D scan with image structure.

Windowed or quadtree-style paths improve locality [13, 28], yet they are still constrained by predefined geometric templates. As a result, the sequence fed to the SSM is only weakly aligned with the semantic structure of the image. This is especially problematic when object boundaries are curved, elongated, or span distant yet semantically related pixels.

The key idea of this work is to make the scan itself deformable. Instead of deciding only how to process a fixed token sequence, we also learn how to construct that sequence from the feature map. Inspired by deformable sampling in CNNs and Transformers [2, 25, 34, 35], we introduce a deformable scanning mechanism that jointly learns two complementary transformations: spatial point offsets that move sampling locations toward informative regions, and token-index offsets that reorder the resulting tokens into a content-adaptive path. The first addresses *where* to read features; the second addresses *how* to traverse them.

Based on this idea, we propose DefMamba, a hierarchi-

cal backbone composed of Deformable Mamba blocks (DM blocks). Each block replaces the conventional scan operator with a Deformable State Space Model (DSSM). The DSSM contains three branches: a forward scan, a backward scan, and a deformable scan. The deformable branch predicts bounded offsets using a lightweight offset network with depthwise convolution, channel attention, normalization, and a 1×1 projection. Point offsets are applied by bilinear sampling with a relative offset bias, while token-index offsets are converted into a permutation through differentiable sorting. The outputs of the three branches are aligned and averaged, which preserves the robustness of standard bidirectional scanning while adding content adaptivity.

Experiments show that the approach is effective and practical. On ImageNet-1K, DefMamba-T improves over PlainMamba-L1 by 0.7 points while using substantially fewer FLOPs (1.2G vs. 3.0G). On COCO with Mask R-CNN, DefMamba-S slightly outperforms VMamba-T in both box AP and mask AP with lower backbone FLOPs. On ADE20K with UPerNet, DefMamba-S reaches the best single-scale and multi-scale mIoU among the compared SSM backbones in Table 3. Ablations confirm that deformable points and deformable token ordering are complementary, and branch ablations show that the deformable scan is most effective when paired with standard forward/backward scans.

Our contributions are three-fold. First, we introduce a deformable scanning formulation for visual SSMs that jointly adapts sampling positions and token order. Second, we instantiate this idea as the DSSM, including bounded offsets, relative offset bias, and a differentiable approximation for sorting-based token reordering. Third, we build a practical hierarchical backbone that delivers strong accuracy-efficiency trade-offs on classification, detection, and segmentation.

2. Related Work

2.1. Visual State Space Models

SSMs have become a strong alternative to attention for long-context sequence modeling [6, 7]. Their extension to vision has produced both plain and hierarchical backbones. ViM [33] applies bidirectional Mamba to patch sequences, while VMamba [15] introduces 2D selective scan to recover directional spatial context. Later work improves scan design in different ways: LocalMamba constrains scanning within windows [13], EfficientVMamba uses atrous selective scan [19], MSVMamba adds hierarchical multi-scale scanning [21], GrootVL builds dynamic tree topologies [27], and QuadMamba learns quadtree-based partitions [28]. These methods demonstrate that scan design is crucial, but most still rely on pre-specified routing rules or discrete partitions. DefMamba instead learns continuous sampling locations and continuous token-order scores directly from content.

2.2. Content-Adaptive Visual Sampling

Learning where to sample has a long history in vision. Deformable convolution augments fixed convolution grids with learned offsets [2, 34], and later variants improve efficiency and effectiveness [29]. In Transformers, deformable attention [25, 35] replaces dense or fixed sparse neighborhoods with data-dependent sampling locations. Deformable patch tokenization and progressive token sampling have also been studied for visual recognition [1, 31]. Our method is related in spirit, but the goal is different: we do not replace convolution or attention with deformable operators. Instead, we use deformable sampling to construct a better 1D sequence for a selective SSM. This makes deformability operate at the interface between 2D feature maps and 1D scan dynamics.

2.3. Sequence Construction for Visual Backbones

Sequence order strongly affects what a sequential model can represent. CNNs encode locality through fixed receptive fields [10, 17], while Vision Transformers and hierarchical Transformers rely on tokenization and attention structure [3, 5, 9, 12, 16, 22, 24]. Visual SSMs inherit a related issue because flattening order determines which tokens become adjacent in the recurrence. Existing scan redesigns improve locality or hierarchy, but they typically choose one ordering rule for all images or layers. DefMamba addresses the remaining gap by predicting an image-conditioned order for each feature map, while still keeping the efficient selective scan core.

3. Method

3.1. Overview

Given an input image, DefMamba builds a four-stage hierarchical backbone as shown in Figure 2. Each stage contains several DM blocks followed by spatial downsampling, producing multi-scale features for classification and dense prediction. The tiny, small, and base variants use stage depths $(2, 2, 5, 2)$, $(2, 2, 6, 2)$, and $(2, 3, 16, 2)$, with embedding dimensions 48, 96, and 96 in the first stage, respectively. The core operator inside each DM block is the proposed DSSM.

For an intermediate feature map $X \in \mathbb{R}^{B \times H \times W \times C}$, the DSSM first projects channels, applies a depthwise convolution and SiLU activation, and then constructs three scan branches:

$$\mathcal{S}(X) = \{s_{\text{fwd}}(X), s_{\text{bwd}}(X), s_{\text{def}}(X)\}, \tag{1}$$

where the first two correspond to the standard forward and reverse raster sequences, and the third is a deformable

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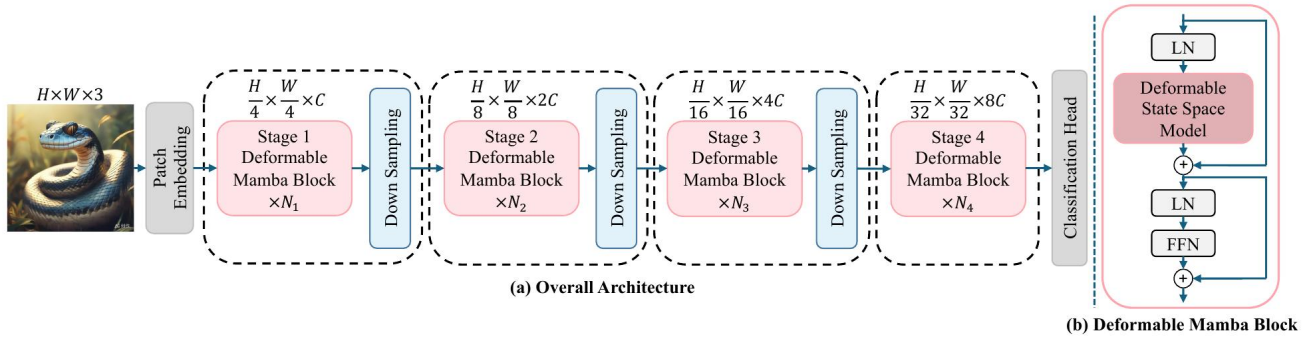

Figure 2. Overview of DefMamba. Left: the four-stage hierarchical backbone. Right: each Deformable Mamba block applies layer normalization, a Deformable State Space Model, and an FFN with residual connections.

content-adaptive sequence. Each branch is processed by a selective SSM with shared overall structure but separate directional states. After scan processing, the backward branch is flipped back to the original order and the deformable branch is inverse-permuted to the spatial layout. The three outputs are averaged before the final projection:

$$Y = \frac{1}{3}\Big(Y_{\text{fwd}} + \text{rev}(Y_{\text{bwd}}) + \pi^{-1}(Y_{\text{def}})\Big). \tag{2}$$

This simple aggregation is important in practice: the released implementation shows that the deformable branch alone is not the strongest configuration, while coupling it with the standard forward/backward pair yields the best result.

3.2. Deformable Scanning

The deformable branch learns both spatial sampling locations and scan order. Figure 3 illustrates the process.

Offset network. We use a lightweight predictor built from a stage-dependent depthwise convolution, channel attention, layer normalization, GELU, and a 1×1 convolution. The depthwise kernel sizes are $(9, 7, 5, 3)$ from shallow to deep stages, enlarging the receptive field early and reducing cost later. For a feature map in channel-first form $\tilde{X} \in \mathbb{R}^{B\times C\times H\times W}$, the predictor outputs three channels:

$$O = f_{\text{off}}(\tilde{X}) \in \mathbb{R}^{B\times 3\times H\times W}. \tag{3}$$

We split them into 2D point offsets $\Delta p \in \mathbb{R}^{B\times 2\times H\times W}$ and a scalar token-index offset $\Delta q \in \mathbb{R}^{B\times 1\times H\times W}$.

Deformable points. Let p_{ij}^{ref} be the normalized reference coordinate of location (i, j). We bound offsets with $\tanh$ and normalize them by spatial size:

$$\hat{\Delta p}_{ij} = \tanh(\Delta p_{ij}) \odot \Big[\frac{1}{H-1}, \frac{1}{W-1}\Big]. \tag{4}$$

The deformable sampling point is

$$p_{ij} = p_{ij}^{\text{ref}} + \Delta p_{ij}. \tag{5}$$

Features are sampled from $\tilde{X}$ by bilinear interpolation at p_{ij}, implemented with `grid_sample`. To preserve positional consistency after moving points, DefMamba adds a learned relative offset bias. Concretely, a small learnable bias table is resized to the current resolution and sampled using the displacement between the key-grid coordinates and the deformable points. The final sampled feature map is the sum of the bilinear feature sample and this position-dependent bias.

Deformable token index. Point deformation changes where features come from, but it does not change the sequence order. We therefore also learn a deformable token path. Let $q^{\text{ref}} \in \mathbb{R}^N$ be the raster index reference for $N = HW$ tokens after normalization to $[-1, 1]$. The predictor outputs

$$\hat{\Delta q} = \tanh(\Delta q), \qquad q = q^{\text{ref}} + \hat{\Delta q}. \tag{6}$$

Sorting q gives a permutation $\pi = \text{argsort}(q)$, which defines the scan order. The sampled tokens are gathered according to π and passed to the deformable SSM branch. Because sorting is discrete, the implementation uses a custom autograd function that routes gradients through the gathered sequence and uses an approximation for the order scores, enabling end-to-end training.

3.3. Deformable State Space Model

The complete DSSM combines the deformable scan with conventional bidirectional scanning. In implementation, the three input sequences are

$$\{x,\ \text{flip}(x),\ \text{gather}(x^{\text{def}}, \pi)\}, \tag{7}$$

where x is the flattened raster sequence and x^{def} is the bilinearly sampled feature map. These three sequences are

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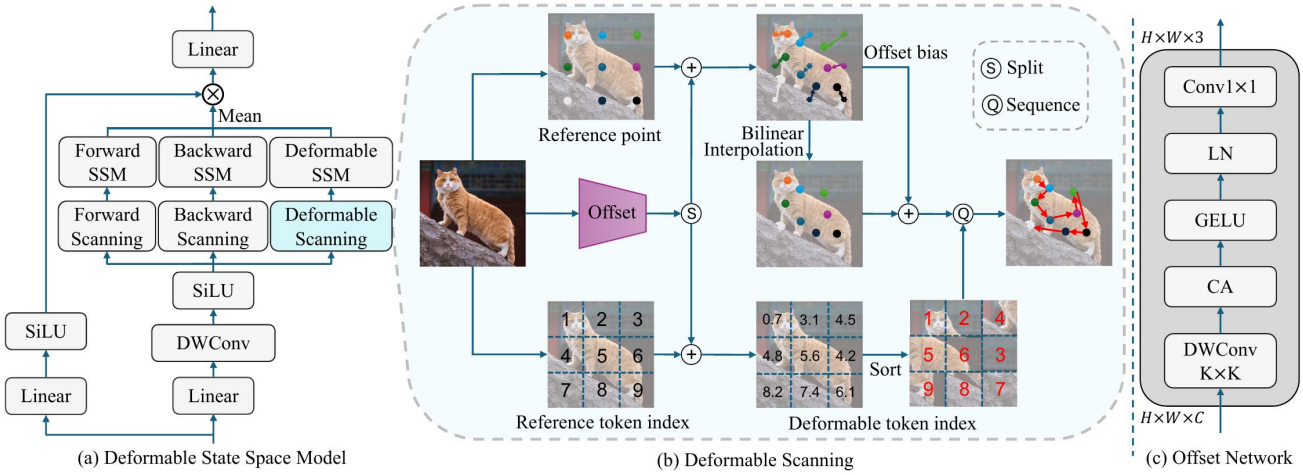

Figure 3. Detailed view of the proposed DSSM. The operator combines forward, backward, and deformable scan branches. The deformable branch predicts spatial point offsets for bilinear sampling and token-index offsets for content-adaptive reordering, implemented by sorting learned order scores.

processed in parallel by the selective scan operator, and the deformable output is inverse-permuted back to the raster order before fusion.

This design addresses two practical issues. First, point deformation and dynamic reordering inject stronger content adaptivity than a fixed route, which improves object-aware representation learning. Second, the forward and backward branches provide a stable optimization anchor. The branch ablation in Table 5 shows that a deformable branch alone reaches 76.5% top-1 on the tiny model, below the forward-backward baseline at 76.9%, but combining them improves accuracy to 78.6%.

3.4. Backbone Instantiation

Each DM block follows a pre-norm residual design:

$$X \leftarrow X + \mathrm{DSSM}(\mathrm{LN}(X)), \qquad X \leftarrow X + \mathrm{MLP}(\mathrm{LN}(X)). \tag{8}$$

The MLP expansion ratio is 4.0 in all released classification models. The backbone uses a convolutional patch embedding stem and strided convolutional downsampling between stages. For downstream tasks, the same backbone is exposed through MMDetection and MMSegmentation wrappers, producing the four stage outputs required by FPN and UPerNet.

4. Experiments

4.1. Experimental Setup

Classification. We evaluate on ImageNet-1K [4] and report top-1 accuracy on the validation set. Following the released classification configuration, DefMamba is trained for 300 epochs with AdamW [18], cosine learning-rate decay, label smoothing, stochastic depth, Mixup, CutMix, and random erasing. Input images are resized to 224×224.

Detection and segmentation. For object detection and instance segmentation, we use Mask R-CNN [11] with a $1\times$ schedule on MS COCO 2017 [14]. For semantic segmentation, we use UPerNet [26] on ADE20K [32] with crop size 512^2 and the standard 160K training schedule inherited from the released MMSegmentation configuration. In both cases, we replace the backbone in the corresponding Swin-based configuration with DefMamba-S and keep the rest of the task framework unchanged.

Model variants. The three released classification variants are DefMamba-T, DefMamba-S, and DefMamba-B. Their parameter/FLOP counts are reported directly in Table 1. Unless noted otherwise, analysis experiments use the tiny setting because it is inexpensive and highlights relative changes clearly.

4.2. Main Results on ImageNet

Table 1 compares DefMamba against CNN, Transformer, and SSM backbones. DefMamba-T reaches 78.6% top-1 with 8M parameters and 1.2G FLOPs, improving over PlainMamba-L1 by 0.7 points while reducing FLOPs from 3.0G to 1.2G. It also outperforms other lightweight SSM baselines such as ViM-T, EfficientVMamba-T, and MSVMamba-N.

At medium scale, DefMamba-S attains 83.5% top-1 at 4.8G FLOPs, slightly surpassing GrootV-T and outperforming LocalVMamba-T, EfficientVMamba-B, MSVMamba-T, and PlainMamba-L2 at comparable or lower cost. At

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Method	Type	#Param.	#FLOPs	Top-1 Acc.
RegNetY-800M [20]	C	6M	0.8G	76.3
GhostNet 1.3× [8]	C	7M	0.2G	75.7
DeiT-Ti [22]	T	6M	1.3G	72.2
ViM-T [33]	S	7M	1.5G	76.1
EffVMamba-T [19]	S	6M	0.8G	76.5
LocalVim-T [13]	S	8M	1.5G	76.2
PlainMamba-L1 [30]	S	7M	3.0G	77.9
MSVMamba-N [21]	S	7M	0.9G	77.3
QuadMamba-Li [28]	S	5M	0.8G	74.2
DefMamba-T	S	8M	1.2G	**78.6**
RegNetY-4G [20]	C	21M	4.0G	80.0
ConvNeXt-T [17]	C	29M	4.5G	82.1
Conv2Former-T [12]	C	27M	4.4G	83.2
DeiT-S [22]	T	22M	4.6G	79.8
Swin-T [16]	T	29M	4.5G	81.3
CoAtNet-0 [3]	T	25M	4.0G	81.6
CrossFormer-S [24]	T	31M	4.9G	82.5
ViM-S [33]	S	26M	5.1G	81.0
VMamba-T [15]	S	22M	5.6G	82.2

Method	Type	#Param.	#FLOPs	Top-1 Acc.
LocalVim-S [13]	S	28M	4.8G	81.2
LocalVMamba-T [13]	S	26M	5.7G	82.7
EffVMamba-B [19]	S	33M	4.6G	83.0
MSVMamba-T [21]	S	33M	4.6G	82.8
PlainMamba-L2 [30]	S	25M	8.1G	81.6
GrootV-T [27]	S	30M	4.8G	83.4
QuadMamba-S [28]	S	31M	5.5G	82.4
DefMamba-S	S	32M	4.8G	**83.5**
ConvNeXt-S [17]	C	50M	8.7G	83.1
Conv2Former [12]	C	50M	8.7G	84.1
Swin-S [16]	T	50M	8.7G	83.0
CoAtNet-1 [3]	T	42M	8.0G	83.3
CrossFormer-B [24]	T	52M	9.2G	83.4
VMamba-S [15]	S	50M	8.7G	83.6
LocalVMamba-B [13]	S	50M	11.4G	83.7
PlainMamba-L3 [30]	S	50M	14.4G	82.3
GrootV-S [27]	S	51M	8.5G	84.2
QuadMamba-B [28]	S	50M	9.3G	83.8
DefMamba-B	S	51M	8.5G	**84.2**

Table 1. Image classification performance on ImageNet-1K validation set. C, T and S indicate the model type of CNNs, Transformer and SSM. The best results are shown in **bold** font.

Mask R-CNN 1× schedule							
Backbone	**#FLOPs**	**AP^b**	**AP^b_{50}**	**AP^b_{75}**	**AP^m**	**AP^m_{50}**	**AP^m_{75}**
ResNet-50 [10]	260G	38.2	58.8	41.4	34.7	55.7	37.2
Swin-T [16]	267G	42.7	65.2	46.8	39.3	62.2	42.2
ConvNeXt-T [17]	262G	44.2	66.6	48.3	40.1	63.3	42.8
PVTv2-B2 [23]	309G	45.3	67.1	49.4	41.2	64.2	44.4
VMamba-T [15]	286G	47.4	69.5	52.0	42.7	66.3	46.0
LocalVMamba-T [13]	291G	46.7	68.7	50.8	42.2	65.7	45.5
QuadMamba-S [28]	301G	46.7	69.0	51.3	42.4	65.9	45.6
MSVMamba-T [21]	252G	46.9	68.8	51.4	42.2	65.6	45.4
GrootV-T [27]	265G	47.0	69.4	51.5	42.7	66.4	46.0
DefMamba-S	268G	47.5	69.6	51.7	42.8	66.3	46.2

Table 2. Object detection and instance segmentation performance on MSCOCO 2017 val set. All using Mask R-CNN framework. AP^b and AP^m indicate the mean Average Precision (mAP) of detection and segmentation, respectively

ADE20K with crop size 512				
Backbone	**mIOU (SS)**	**mIOU (MS)**	**#Param.**	**#FLOPs**
ResNet-50 [10]	42.1	42.8	67M	953G
Swin-T [16]	44.5	45.8	60M	945G
ConvNeXt-T [17]	46.0	46.7	60M	939G
NAT-T [9]	47.1	48.4	58M	934G
Vim-S [33]	44.9	-	46M	-
LocalVim-S [13]	46.4	47.5	58M	297G
VMamba-T [15]	47.9	48.8	62M	949G
PlainMamba-L2 [30]	46.8	-	55M	285G
LocalVMamba-T [13]	47.9	49.1	57M	970G
MSVMamba-T [21]	47.6	48.5	65M	942G
QuadMamba-S [28]	47.2	48.1	62M	961G
GrootV-T [27]	48.5	49.4	60M	941G
DefMamba-S	48.8	49.6	65M	946G

Table 3. Semantic segmentation performance on ADE20K val set. The crop size is all set to 512^2. SS and MS denote single-scale and multi-scale testing, respectively.

larger scale, DefMamba-B reaches 84.2%, matching the best reported number in the comparison table while remaining more efficient than several competing SSM and Transformer baselines. These results indicate that deformable scan construction improves the accuracy-efficiency trade-off across model sizes.

4.3. Transfer to Detection and Segmentation

Table 2 shows that DefMamba-S transfers well to detection and instance segmentation. With Mask R-CNN, it achieves 47.5 AP^b and 42.8 AP^m, slightly outperforming VMamba-T (47.4/42.7) while using fewer FLOPs (268G vs. 286G). The gains are modest but consistent with the idea that a structure-aware scan is useful beyond classification.

On ADE20K in Table 3, DefMamba-S reaches 48.8 mIoU under single-scale testing and 49.6 under multi-scale testing. This exceeds the compared SSM baselines in the table, including VMamba-T, LocalVMamba-T, MSVMamba-T, QuadMamba-S, and GrootV-T. The result suggests that the learned scan is especially beneficial for dense prediction, where preserving object structure and long-range layout both matter.

Task sensitivity. The margins on ImageNet are steady but moderate, whereas the improvement on ADE20K is clearer. Since classification can often tolerate partial disruption of spatial order after global pooling, it benefits mainly from

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Index	DP	DT	OB	CA	#Param.	#FLOPs	Top-1
(1)					7.5M	1.1G	77.0
(2)	√				7.5M	1.2G	77.4
(3)		√			7.5M	1.2G	77.2
(4)	√	√			7.5M	1.2G	77.9
(5)	√	√	√		8.2M	1.2G	78.2
(6)	√	√	√	√	8.3M	1.2G	**78.6**

Table 4. Comparisons with the different components of proposed deformable scanning in tiny model size settings on ImageNet-1k. The best results are shown in **bold** font.

better token interactions. Semantic segmentation is less forgiving: errors in local adjacency and boundary continuity directly affect per-pixel labels. The stronger ADE20K result therefore matches the mechanism of DefMamba, whose deformable points and token order explicitly target object-aligned spatial organization before sequence modeling.

This cross-task pattern also helps interpret the COCO results. Detection and instance segmentation require region-level localization but still aggregate evidence over proposals, so they benefit from cleaner object-aware context without depending as directly on every local boundary decision. ADE20K is stricter: label assignment is dense, class vocabularies are broad, and small boundary errors are penalized immediately in mIoU. The fact that DefMamba yields its clearest advantage there is therefore consistent with the intended role of deformable sequence construction, namely reducing the mismatch between 2D scene structure and the 1D order consumed by the SSM.

5. Analysis

5.1. Ablation on Deformable Components

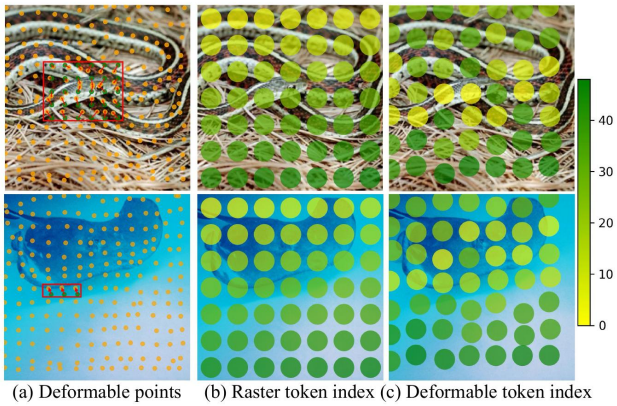
Table 4 isolates the components of deformable scanning in the tiny model. Starting from a 77.0% baseline without deformable components, adding deformable points (DP) alone improves accuracy to 77.4%, while deformable token ordering (DT) alone gives 77.2%. Using both together reaches 77.9%, showing that the two mechanisms are complementary. Adding the offset bias (OB) further improves performance to 78.2%, and channel attention (CA) lifts the final model to 78.6%. Overall, the full design gains 1.6 points over the baseline in the same FLOP regime.

5.2. Ablation on Scanning Branches

Table 5 studies how the deformable branch interacts with other scan branches. The deformable branch by itself is not sufficient: it scores 76.5%, below the forward-backward baseline at 76.9%. Adding previously proposed continuous or local branches helps somewhat, but the best result comes from combining the deformable branch with the standard forward/backward pair, reaching 78.6%. This supports our design choice to preserve bidirectional raster branches for optimization stability rather than replacing them outright.

Index	FB-BB	CB [30]	LB [13]	DB	#Param.	#FLOPs	Top-1
(1)	√				7.3M	1.1G	76.9
(2)				√	8.1M	1.1G	76.5
(3)	√	√			7.5M	1.1G	77.3
(4)	√		√		7.5M	1.1G	77.1
(5)	√			√	8.3M	1.2G	**78.6**

Table 5. Comparisons with the different scanning branches in the setting of tiny model size. DB means our deformable branch. The best results are shown in **bold** font.



(a) Deformable points (b) Raster token index (c) Deformable token index

Figure 4. Visualization of learned deformable points and token order. The learned points move toward salient object regions, and the deformable token index produces an order that better follows object structure than raster indexing.

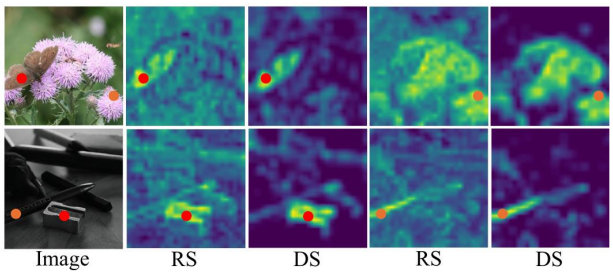


Figure 5. Activation maps at selected query positions. Compared with raster scanning (RS), deformable scanning (DS) produces more object-focused responses around the queried semantic region.

5.3. Qualitative Behavior

Figure 4 visualizes the learned deformable points and token order. The point offsets move reference positions toward object interiors and boundaries instead of remaining on a rigid grid. The token-order visualization further shows that the learned sequence departs from the raster layout most strongly in semantically important regions, which

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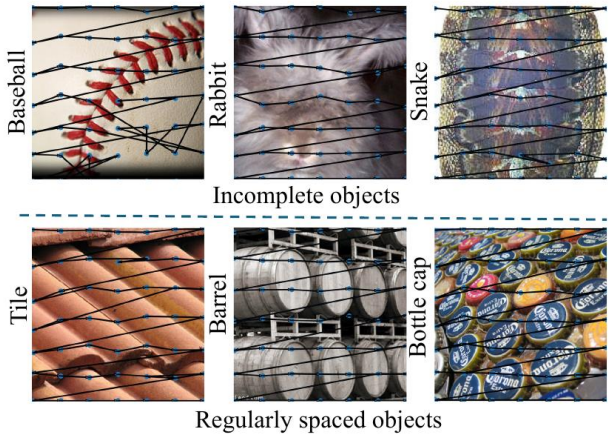

Figure 6. Failure cases. DefMamba still struggles when object structure is incomplete or when many instances form highly regular repeated patterns, which can make the learned path ambiguous.

is consistent with the intended behavior of the deformable path.

Figure 5 compares activation maps for raster scanning and deformable scanning at matched spatial queries. Deformable scanning yields responses that are more concentrated on semantically relevant regions, such as the butterfly body or the metallic object, whereas raster scanning spreads activation more diffusely into the background. This supports the claim that the proposed sequence construction helps the SSM aggregate context along object-aligned paths.

Complementarity. The gains in Table 4 are additive, not redundant. Deformable points refine *what* is sampled around a location, while deformable token order changes *which* regions become adjacent in the recurrence. The activation maps indicate that both effects are needed to form sharper, semantically coherent context propagation.

5.4. Failure Cases

Despite its gains, DefMamba is not universally reliable. Figure 6 shows two recurring failure modes. The first appears on incomplete or partially visible objects, where the learned path may connect only a subset of the true structure. The second occurs for regularly repeated patterns such as tiles, barrels, or bottle caps, where many local alternatives look equally plausible and the scan path becomes less discriminative. These examples suggest that deformable sequence construction improves content awareness but does not fully solve structural ambiguity.

These cases also clarify why the final model keeps bidirectional raster branches. When the adaptive path is ambiguous, the fixed scans still provide full coverage and a stable order. The hybrid design is therefore more accurate on average in Table 5 and less brittle than using a single learned route.

6. Conclusion

We presented DefMamba, a visual state space backbone that replaces rigid scan construction with a deformable alternative. The proposed DSSM learns both spatial sampling points and token order, while retaining forward and backward raster branches to stabilize training. Across ImageNet-1K, COCO, and ADE20K, the method provides a consistently strong accuracy-efficiency trade-off, with especially clear gains in the lightweight regime and on dense prediction tasks. The results indicate that improving how a visual SSM sees the image sequence is a practical way to strengthen representation quality without abandoning the efficiency advantages of selective scan.

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References

[1] Zhiyang Chen, Yousong Zhu, Chaoyang Zhao, Guosheng Hu, Wei Zeng, Jinqiao Wang, and Ming Tang. Dpt: Deformable patch-based transformer for visual recognition. In *Proceedings of the 29th ACM international conference on multimedia*, pages 2899–2907, 2021.

[2] Jifeng Dai, Haozhi Qi, Yuwen Xiong, Yi Li, Guodong Zhang, Han Hu, and Yichen Wei. Deformable convolutional networks. In *Proceedings of the IEEE international conference on computer vision*, pages 764–773, 2017.

[3] Zihang Dai, Hanxiao Liu, Quoc V Le, and Mingxing Tan. Coatnet: Marrying convolution and attention for all data sizes. *Advances in neural information processing systems*, 34:3965–3977, 2021.

[4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.

[5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[6] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

[7] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. *arXiv preprint arXiv:2111.00396*, 2021.

[8] Kai Han, Yunhe Wang, Qi Tian, Jianyuan Guo, Chunjing Xu, and Chang Xu. Ghostnet: More features from cheap operations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1580–1589, 2020.

[9] Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6185–6194, 2023.

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[11] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017.

[12] Qibin Hou, Cheng-Ze Lu, Ming-Ming Cheng, and Jiashi Feng. Conv2former: A simple transformer-style convnet for visual recognition. *IEEE transactions on pattern analysis and machine intelligence*, 2024.

[13] Tao Huang, Xiaohuan Pei, Shan You, Fei Wang, Chen Qian, and Chang Xu. Localmamba: Visual state space model with windowed selective scan. *arXiv preprint arXiv:2403.09338*, 2024.

[14] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*, pages 740–755. Springer, 2014.

[15] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, and Yunfan Liu. Vmamba: Visual state space model 2024. *arXiv preprint arXiv:2401.10166*, 2024.

[16] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021.

[17] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022.

[18] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2017.

[19] Xiaohuan Pei, Tao Huang, and Chang Xu. Efficientvmamba: Atrous selective scan for light weight visual mamba. *arXiv preprint arXiv:2403.09977*, 2024.

[20] Ilija Radosavovic, Raj Prateek Kosaraju, Ross Girshick, Kaiming He, and Piotr Dollár. Designing network design spaces. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10428–10436, 2020.

[21] Yuheng Shi, Minjing Dong, and Chang Xu. Multi-scale vmamba: Hierarchy in hierarchy visual state space model. *arXiv preprint arXiv:2405.14174*, 2024.

[22] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning*, pages 10347–10357. PMLR, 2021.

[23] Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and

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Ling Shao. Pvt v2: Improved baselines with pyramid vision transformer. *Computational Visual Media*, 8(3):415–424, 2022.

[24] Wenxiao Wang, Wei Chen, Qibo Qiu, Long Chen, Boxi Wu, Binbin Lin, Xiaofei He, and Wei Liu. Crossformer++: A versatile vision transformer hinging on cross-scale attention. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3123–3136, 2023.

[25] Zhuofan Xia, Xuran Pan, Shiji Song, Li Erran Li, and Gao Huang. Vision transformer with deformable attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4794–4803, 2022.

[26] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *Proceedings of the European conference on computer vision (ECCV)*, pages 418–434, 2018.

[27] Yicheng Xiao, Lin Song, Shaoli Huang, Jiangshan Wang, Siyu Song, Yixiao Ge, Xiu Li, and Ying Shan. Grootvl: Tree topology is all you need in state space model. *arXiv preprint arXiv:2406.02395*, 2024.

[28] Fei Xie, Weijia Zhang, Zhongdao Wang, and Chao Ma. Quadmamba: Learning quadtree-based selective scan for visual state space model. *arXiv preprint arXiv:2410.06806*, 2024.

[29] Yuwen Xiong, Zhiqi Li, Yuntao Chen, Feng Wang, Xizhou Zhu, Jiapeng Luo, Wenhai Wang, Tong Lu, Hongsheng Li, Yu Qiao, et al. Efficient deformable convnets: Rethinking dynamic and sparse operator for vision applications. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5652–5661, 2024.

[30] Chenhongyi Yang, Zehui Chen, Miguel Espinosa, Linus Ericsson, Zhenyu Wang, Jiaming Liu, and Elliot J Crowley. Plainmamba: Improving non-hierarchical mamba in visual recognition. *arXiv preprint arXiv:2403.17695*, 2024.

[31] Xiaoyu Yue, Shuyang Sun, Zhanghui Kuang, Meng Wei, Philip HS Torr, Wayne Zhang, and Dahua Lin. Vision transformer with progressive sampling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 387–396, 2021.

[32] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Scene parsing through ade20k dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 633–641, 2017.

[33] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model. *arXiv preprint arXiv:2401.09417*, 2024.

[34] Xizhou Zhu, Han Hu, Stephen Lin, and Jifeng Dai. Deformable convnets v2: More deformable, better results. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9308–9316, 2019.

[35] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. *arXiv preprint arXiv:2010.04159*, 2020.

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